Is Recourse Equally Open to Everyone?

Auditing algorithmic recourse in public-sector-style eligibility decisions: the 1994 benchmark vs 2023 North Carolina census data

Ogechukwu Ezenwa · AIPI 590: Advanced Topics in AIPI · Duke University · Fall 2026

Code · app · full report

1 What is recourse?

If a model denies you, recourse is a change you could make to get approved.

DENIED
works 40 hours/week

→

APPROVED
works 45 hours/week

An explanation says *why* you were denied. Recourse says *what you can do*. "Be older" or "get married" may flip the model, but it is not recourse: no one can act on it.

2 The question

When tools suggest how to reverse an automated denial, is the path realistic, and equally open to every group? Do findings from the 1994 benchmark still hold in 2023?

3 How the audit works

Data	UCI Adult 1994 (48,842 people) and ACS North Carolina 2023 (56,486 workers). "Approved" = income above $50K in 1994, $103K in 2023 (the same bar in 2023 dollars).
Models	Gradient boosting and logistic regression. Sex and race are not model inputs.
Realistic	Cheapest change within 6 effort units (1 = a year of school), touching only what a person can change. Immutable: age, marital status, household role, birthplace, capital gains. Up only: education. Bounded: hours, work sector, occupation.
Off-the-shelf	Any feature may change, cheapest by plain distance, as generic counterfactual tools do.
Statistics	~800 denied applicants per dataset; 95% bootstrap CIs; groups compared at the same starting distance from approval.

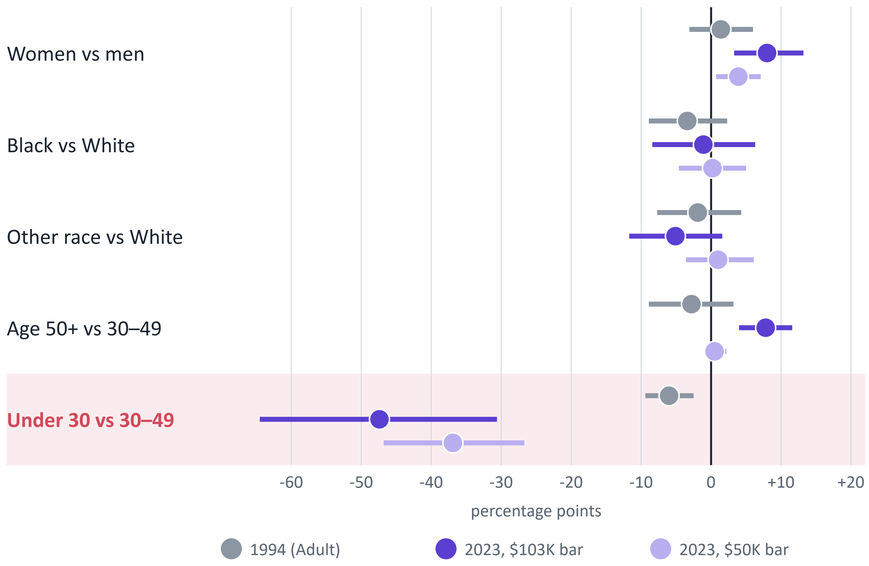
Young applicants have almost no path to approval, and the gap grew

8%
of denied under-30s have realistic recourse (2023)

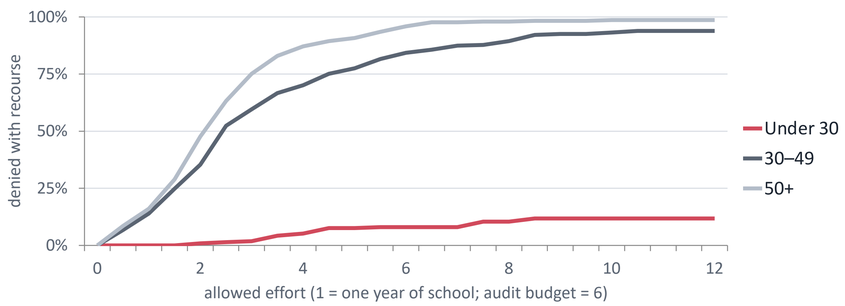
84%
of denied 30–49-year-olds do

−47 pp
gap at the same starting distance (2023), up from −6 pp in 1994

Recourse gap vs the largest group, compared at the same starting distance (95% CI)



More effort does not close the gap (2023)



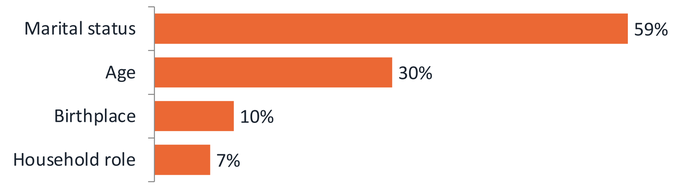
4 Off-the-shelf advice is mostly impossible

95%
of its advice breaks a rule (1994)

77%
with capital gains held fixed

42%
in 2023 data

What it tells people to change (1994, capital gains fixed)



93% of its advice to denied under-30s in 2023 amounts to "be older" (median: be 48).

5 1994's sex and race gaps came from where people start

1994	Raw gap	Same starting distance
Women vs men	−19 pp*	+1 pp
Black vs White	−21 pp*	−3 pp

Denied further from the line, but treated alike at the same distance. In 2023, women have more recourse than men (+8 pp) and the race gap is not significant. * 95% CI excludes 0

6 Why it matters

- Error rates flag age (98% of qualified under-30s are wrongly denied); recourse shows that, once denied, almost none can do anything about it.
- Unconstrained explanations mislead most where real recourse is scarcest.
- Re-audit on current data: the 1994 benchmark understated the age gap.

Limits: income is a proxy label (in benefits, lower income qualifies); costs and immutability are explicit judgments; heuristic search; one state.

References: Wachter et al. 2017 · Ustun et al. 2019 · Gupta et al. 2019 · Mothilal et al. 2020 (DiCE) · Ding et al. 2021 (Retiring Adult) · Karimi et al. 2022 · full list in REPORT.md. Research audit on public survey data, not an eligibility system.